Problem Set 4

(1) What is the shape of research productivity in phuzics over the life cycle?

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Model

- Model estimated for research productivity in phuzics

$$y_{it} = \beta_0 + \sum_{s=1}^{2} \rho_i y_{it-s} + \beta_1 EXP_{it} + \beta_2 {EXP_{it}}^2 + \beta_3 IER_{it} + \beta_4 x_i + \beta_5 PHD80_i + \alpha_i + u_{it}$$

$\boldsymbol{y_{it}}$: $the \log of\ published\ pages\ equivalents\ of\ i^{th}\ individual\ at\ time\ t$
$\boldsymbol{EXP_{it}}$: $the\ experience\ years\ after\ getting\ PhD$
$\boldsymbol{IER_{it}}$: $the\ reciprocal\ of\ experience\ times\ the\ rank\ of\ PhD\ granting\ univ.$
$\boldsymbol{x_i}$: $the\ gender(= 1, if\ \ female)$
$\boldsymbol{PHD80_i}$: $the\ post\ 1980(= 1, if\ person\ i\ got\ PhD\ after\ 1980)$
$\boldsymbol{\alpha_i}$: $the\ individual\ fixed\ effect$

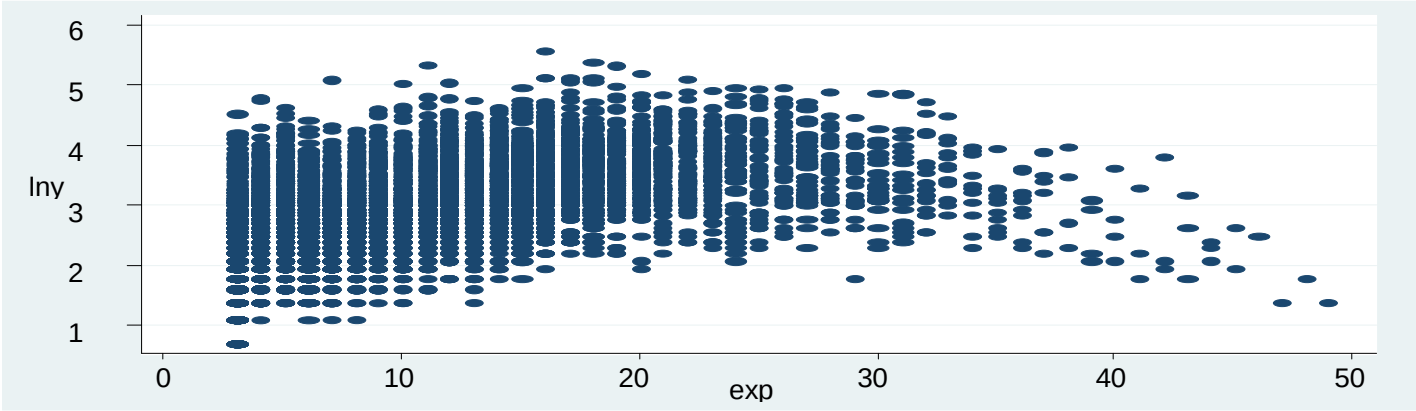
Model

- The model has some characteristics as follows;

(1) The research productivity (y_{it}) is measured by annual number of published pages ($y_{i\,t-s}$).

(2) The shape of research productivity in phuzics over change in experience is the non-linear relationship as in figure below, which implies we need to include as covariate, both of EXP_{it} and quadratic form of it (EXP_{it}^2).

Figure 1. Relationship between log (y) and EXP



Model

(3) The effect of the rank is assumed to fade out over time, thus we can include as a covariate, $IER_{it} \equiv (EXP_{it} \times rPHD_{it})^{-1}$, $where\ rPHD_{it} \equiv$ the rank of PhD granting university).

(4) To catch the effect of gender and whether to exist "the phuzical revolution" of the 1980's (i.e., paradigm shift), we also add as covariates, gender dummy (x_i), and dummy for PhD vintage of post-1980 ($PHD80_i$).

(5) Finally, we include the researcher's individual fixed effect (α_i).

Result of Estimation

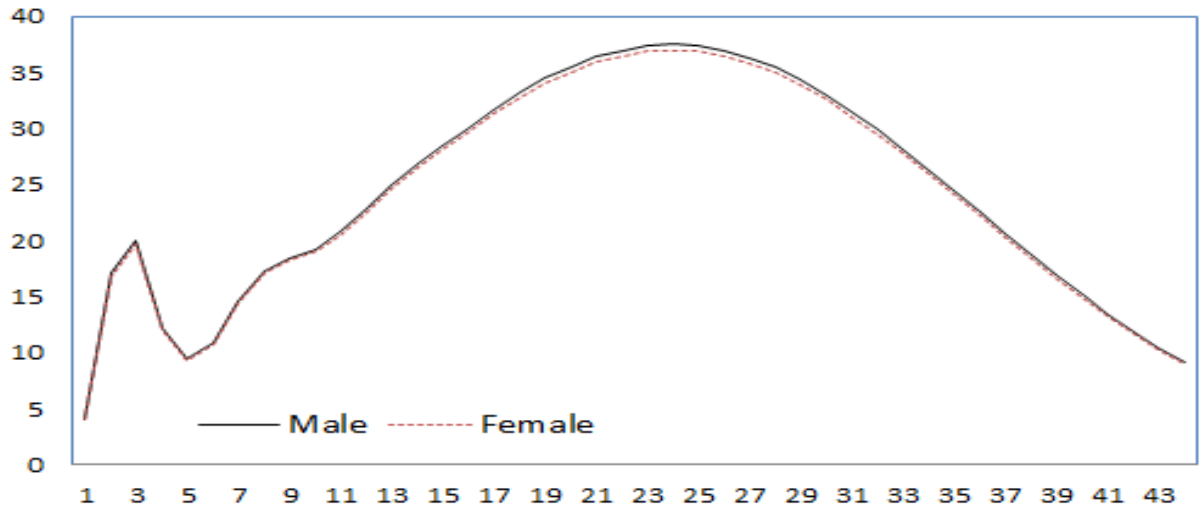
	Pooled OLS	Fixed	Between	Random	Hausman-Taylor
	coef / se	coef / se	coef / se	coef / se	coef / se
y_{it-1}	0.660***	0.543***	1.134***	0.660***	0.543***
	(0.012)	(0.013)	(0.034)	(0.012)	(0.012)
y_{it-2}	-0.347***	-0.426***	-0.393***	-0.347***	-0.426***
	(0.012)	(0.012)	(0.033)	(0.012)	(0.012)
EXP_{it}	0.119***	0.155***	0.070***	0.119***	0.155***
	(0.004)	(0.006)	(0.008)	(0.004)	(0.006)
EXP_{it}^2	-0.002***	-0.003***	-0.002***	-0.002***	-0.003***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
IER_{it}	1.286***	2.183***	0.045	1.286***	2.173***
	(0.190)	(0.447)	(0.186)	(0.190)	(0.439)
$PHD80_i$	0.011	(dropped)	0.031	0.011	0.010
	(0.015)		(0.028)	(0.015)	(0.031)
SEX_i	-0.010	(dropped)	0.008	-0.010	-0.013
	(0.020)		(0.020)	(0.020)	(0.040)
$cons$	1.020***	1.251***	0.222***	1.020***	1.247***
	(0.043)	(0.069)	(0.071)	(0.043)	(0.074)

Note: *** : p-value < 0.01, ** : p-value < 0.05, * : p-value < 0.1

Conclusion

- The results above suggest that a phuzicist's productivity increases after PhD was granted, reaches its maximum at about the 25 years of experience, and then declines after the points.

- We can draw a plot of research productivity of a representative phuzicist who has PhD rank = 2, initial page = 4, PhD80 = 1 for both sexes as figure below.



Conclusion

- The figure above presents that the research productivity goes up until 25 years of experience and falls afterwards for both male and females.